

# Loan Underwriting Model: Algorithmic Fairness & Bias Audit

## 1. Audit Scope & Protected Attributes

Evaluated Random Forest Loan Classifier across 1,000 synthetic test applicant profiles to verify fairness across Gender, Age Groups, and Rural vs Urban geographies.

## 2. Disparate Impact & Fairness Analysis

| Demographic Segment | Approval Rate | False Positive Rate | False Negative Rate | Disparate Impact Ratio     | Status |
|---------------------|---------------|---------------------|---------------------|----------------------------|--------|
| Female Applicants   | 68.4%         | 5.2%                | 4.1%                | 0.98 (Target: 0.80 - 1.25) | PASS   |
| Male Applicants     | 69.8%         | 5.1%                | 4.3%                | 1.00 (Reference Group)     | PASS   |
| Age Group: 18 - 25  | 62.1%         | 6.0%                | 5.8%                | 0.89 (Target: 0.80 - 1.25) | PASS   |
| Age Group: 26 - 55  | 71.3%         | 4.9%                | 3.9%                | 1.02 (Target: 0.80 - 1.25) | PASS   |
| Rural Borrowers     | 66.5%         | 5.4%                | 4.5%                | 0.95 (Target: 0.80 - 1.25) | PASS   |
| Urban Borrowers     | 70.1%         | 5.0%                | 4.0%                | 1.00 (Reference Group)     | PASS   |

## 3. Conclusions & Ethical Standards

- Disparate Impact ratio across all protected classes stays strictly within the 0.80 to 1.25 regulatory threshold.
- Rejection explanations are generated without reference to protected attributes, citing only verified financial indicators such as debt-to-income and repayment history.